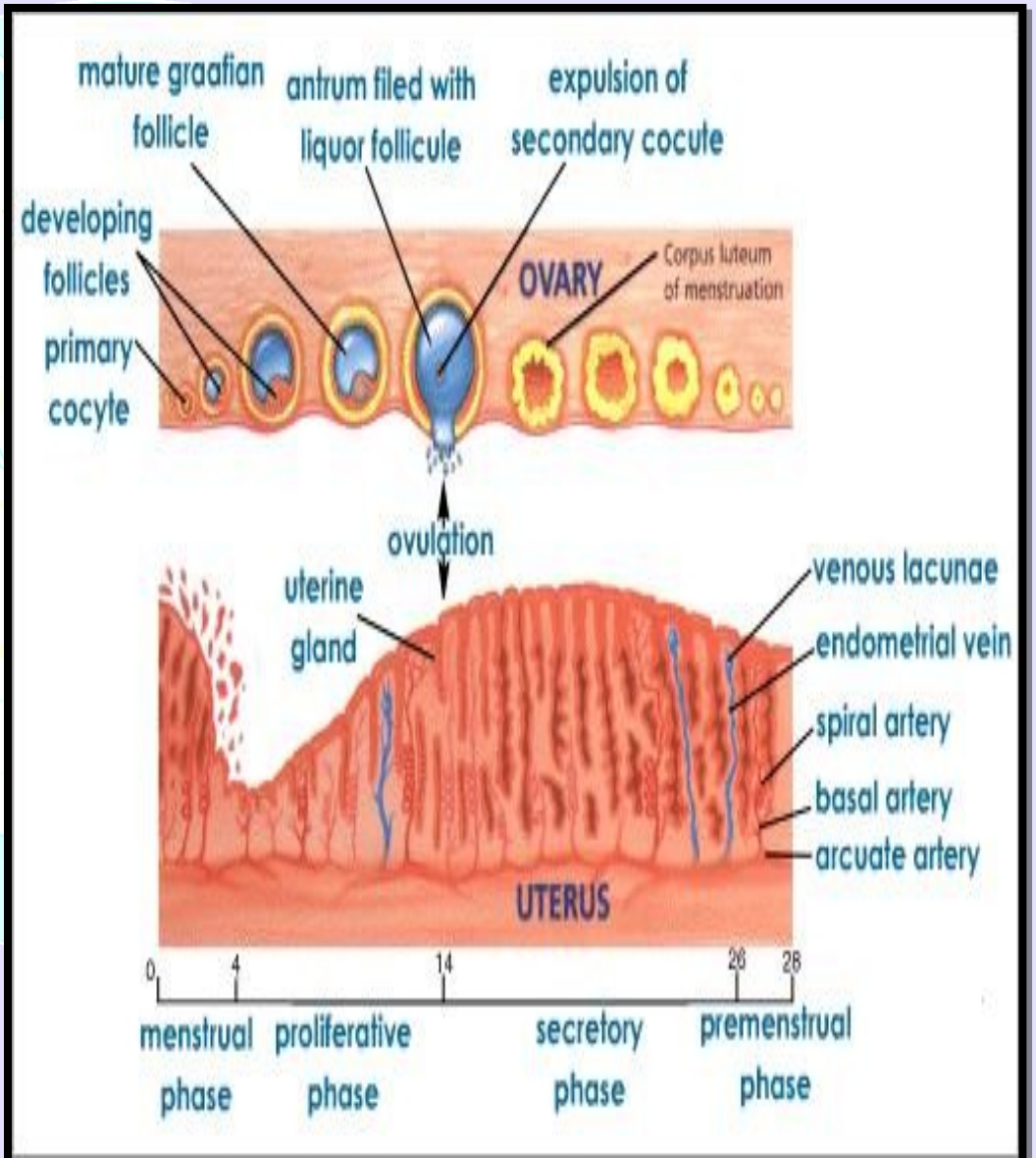
MENSTRUAL CYCLE

LEARNING OBJECTIVES

- **By the completion of the lecture, the student will be able to:**
  - Define the female reproductive cycle(menstrual cycle)
  - Discuss its different phases
  - Interpret the hormonal control of menstrual cycle
  - Correlate the ovarian cycle and with menstrual cycle

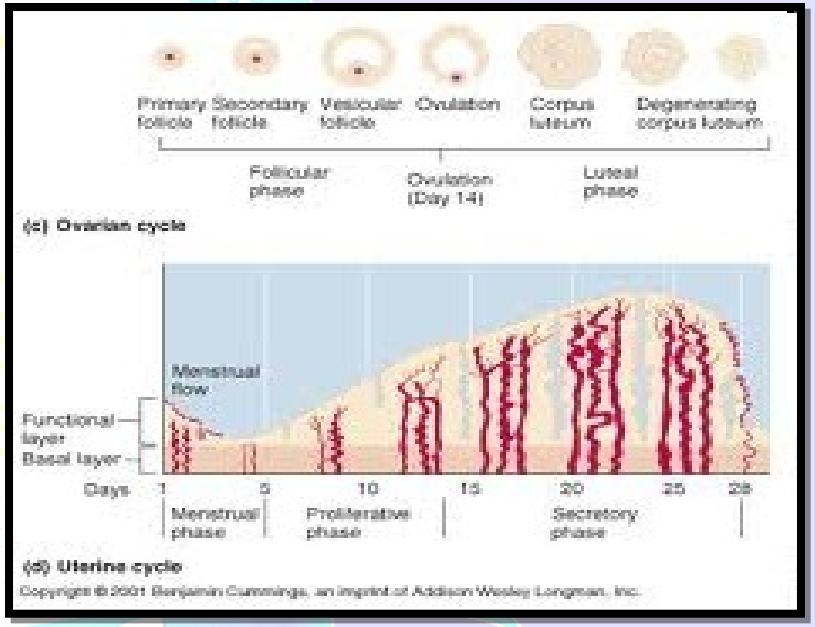
THE MENSTRUAL CYCLE

The menstrual cycle is the periodic vaginal bleeding that occurs with the shedding of the superficial layer Endometrium (menstruation).



MENSTRUATION

- The term menstruation (**menstru = monthly**) is used to indicate periodic shedding of the **stratum functionale** of the endometrium, which becomes thickened prior to menstruation under the control of ovarian steroid hormones.



A : Functional Layer.
B: Basal layer.
C: Myometrium.

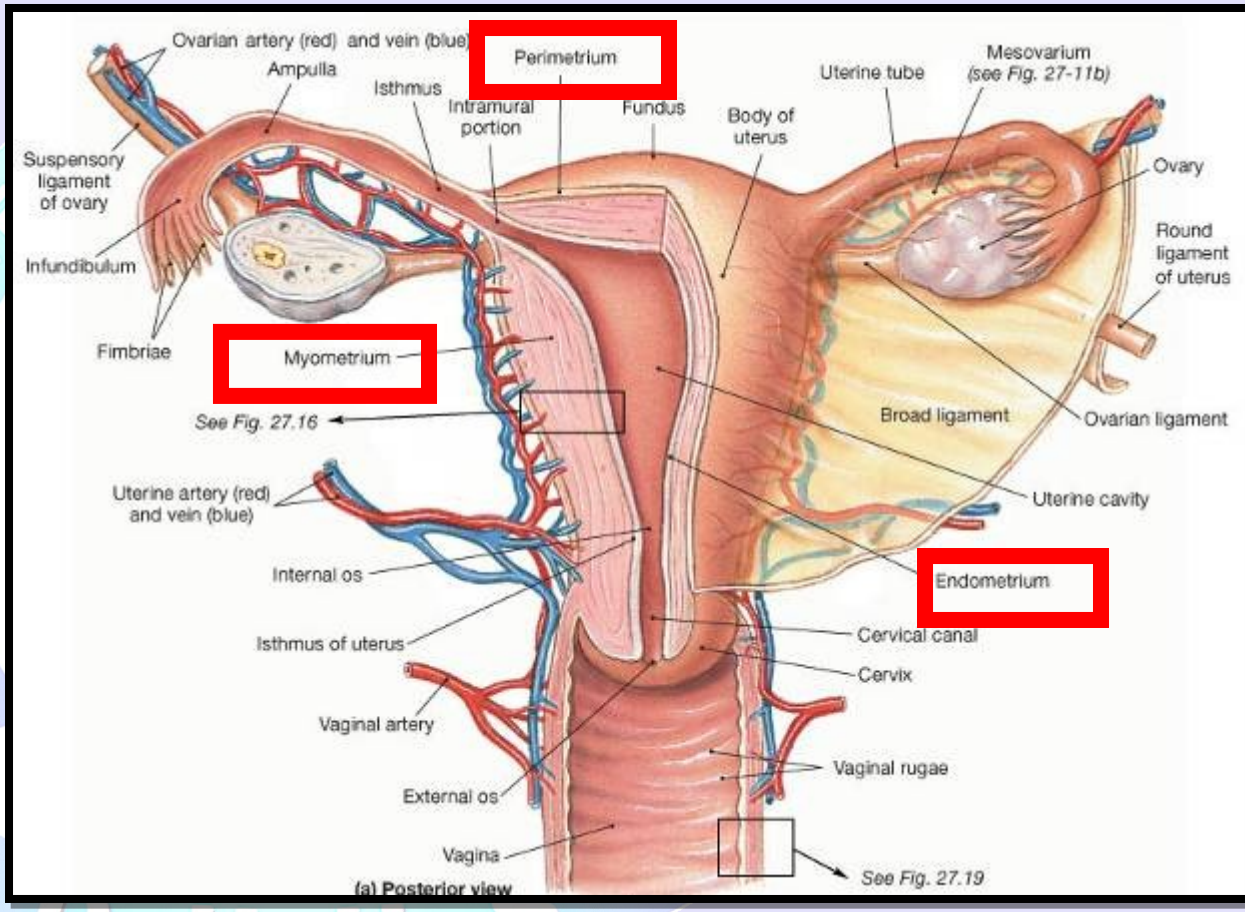
UTERUS

The wall of the uterus consists of three layers:

A) Inner layer Endometrium or mucosa lining the inside wall.

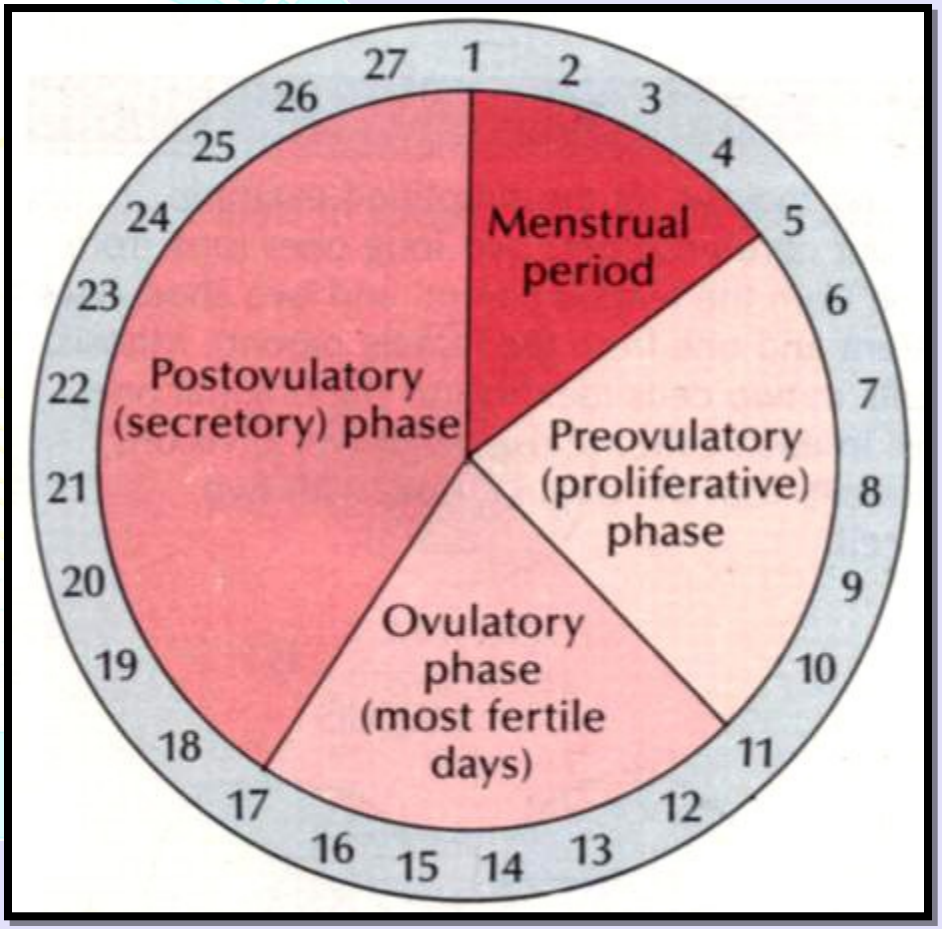
B) Middle layer Myometrium , a thick layer of smooth muscle

C) Outer layer Perimetrium, the peritoneal covering lining the outside wall .

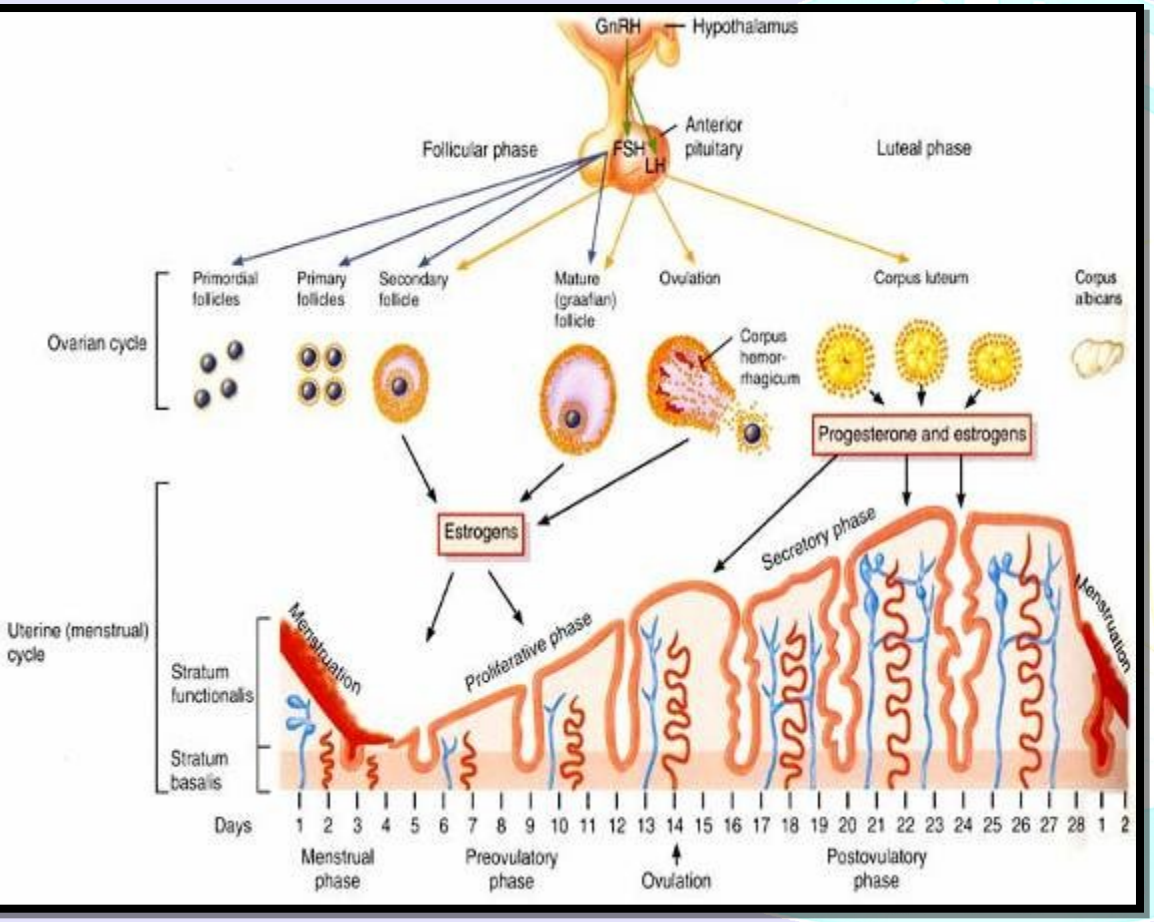


DURATION OF THE CYCLE

- Variable in women
- Typically an average figure is 28 days from the start of one menstrual period to the start of the next.
- It is convenient to call the first day of menstruation “day one” of the cycle.



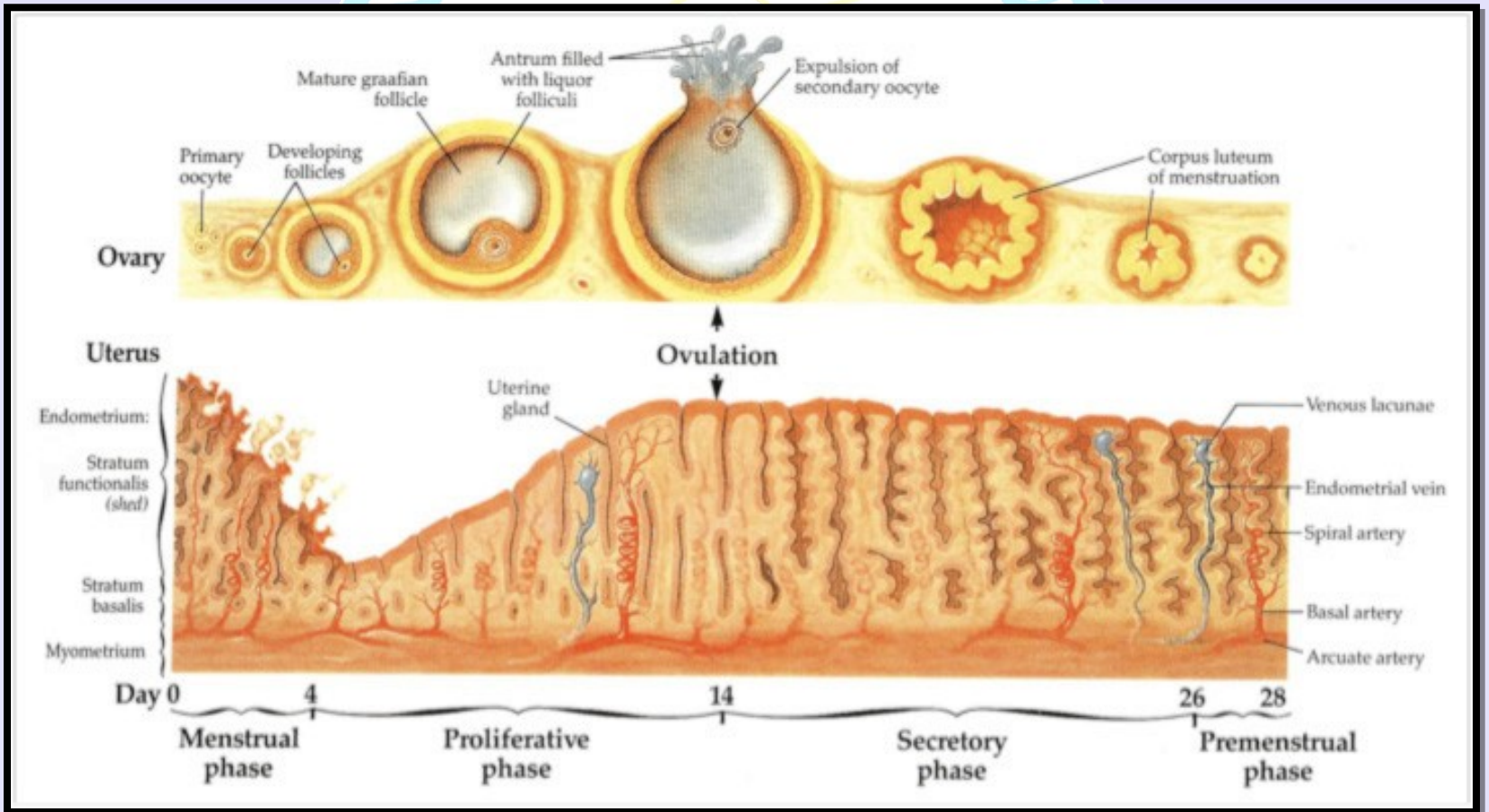
CYCLICAL CHANGES IN THE ENDOMETRIUM



- From puberty (11 to 13 years) untill menopause (45 - 50 years), the endometrium undergoes cyclical changes that occur approximately every 28 days and are under hormonal control by the ovary.

The uterine endometrium passes through three stages during the menstrual cycle:

- follicular or proliferative phase,
- Secretory or progestational phase, and
- Menstrual phase.



LAYERS OF ENDOMETRIUM

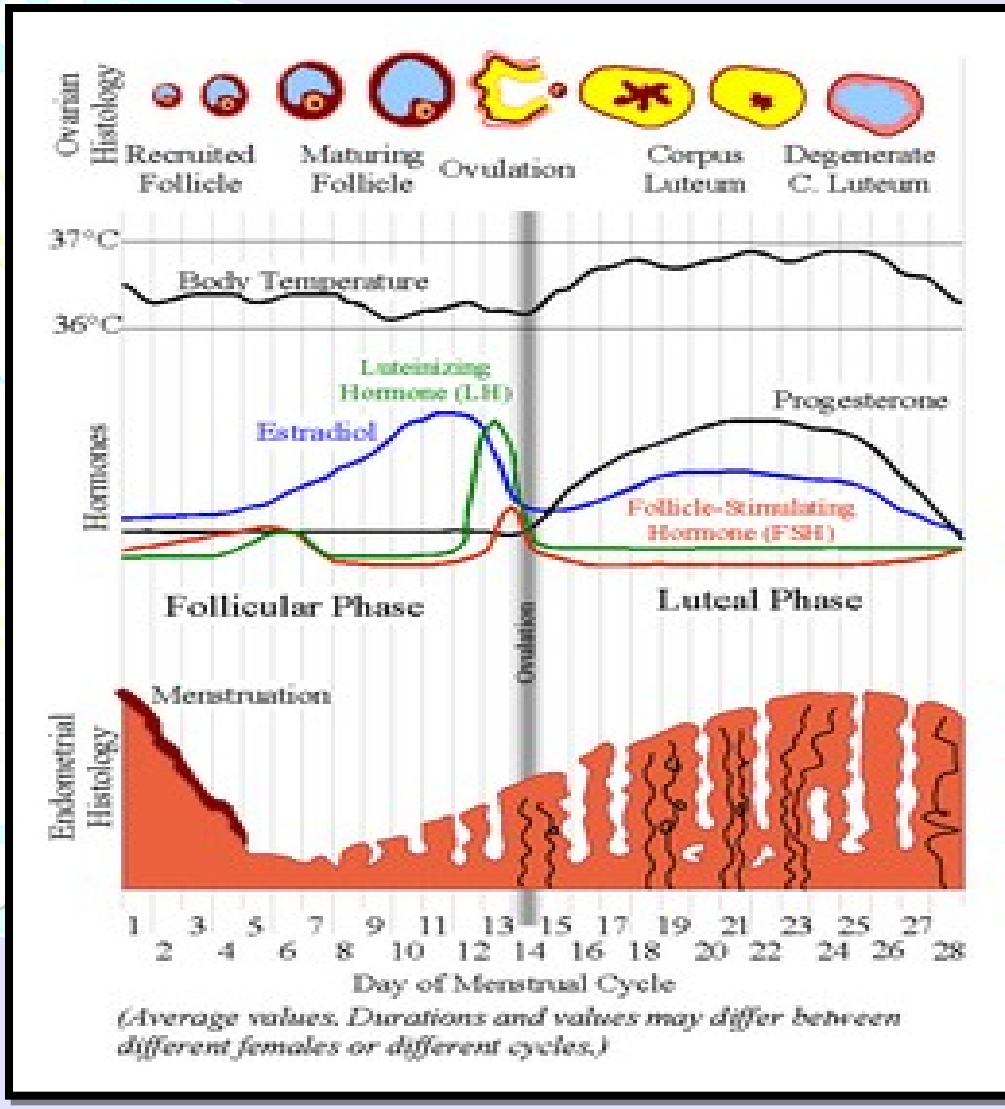
- As a result three distinct layers can be recognized in the endometrium:
  - A superficial compact layer,
  - An intermediate Spongy layer,
  - And a thin basal layer.

REGENERATION OF ENDOMETERIUM

- During the last menstural flow the Stratum Funtionalis shed off and the stratum basalis persists containing the basal (reserve cells).
- During the next cycle the stratum functionalis regenerates under influence of estrogen secreted from the corpus luteum in ovary
- This is the beginning of proliferative phase

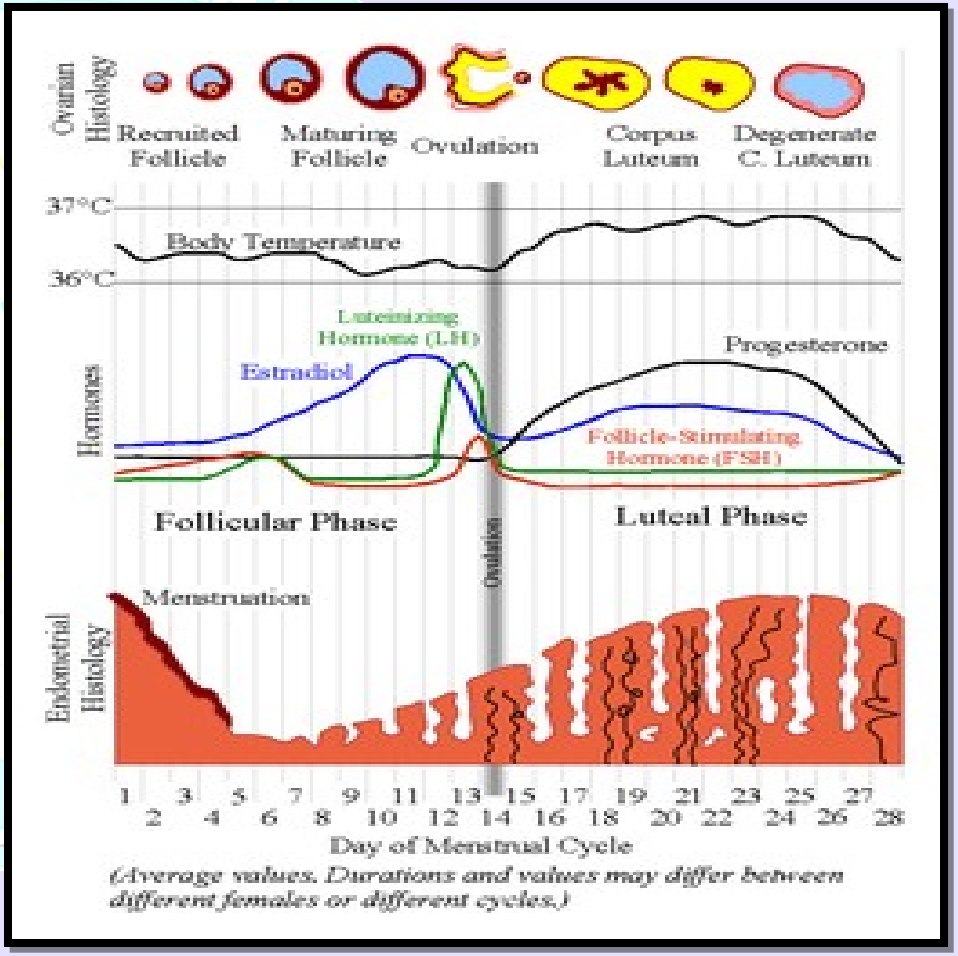
PROLIFERATIVE PHASE

- Last 9 days regeneration of stratum functional
- Formation of new blood vessels which are straight
- Formation of glands which are straight tubular
- This phase under the control of estrogen
- The thickness of stratum functional increases for 2-3 folds



SECRETORY OR PROGESTATIONAL PHASE

- The secretory phase begins approximately 2 – 3 days after ovulation in response to progesterone produced by the corpus luteum.

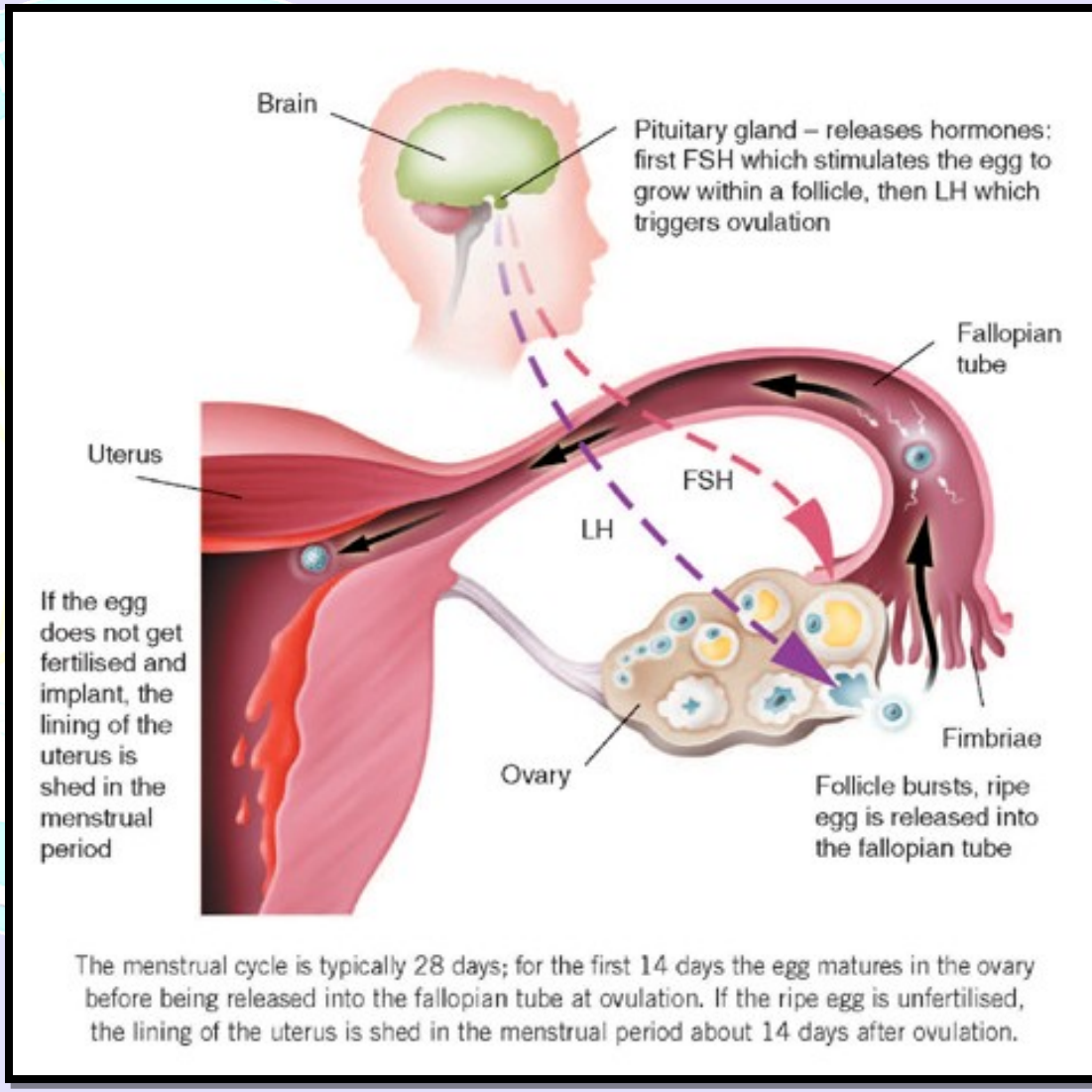


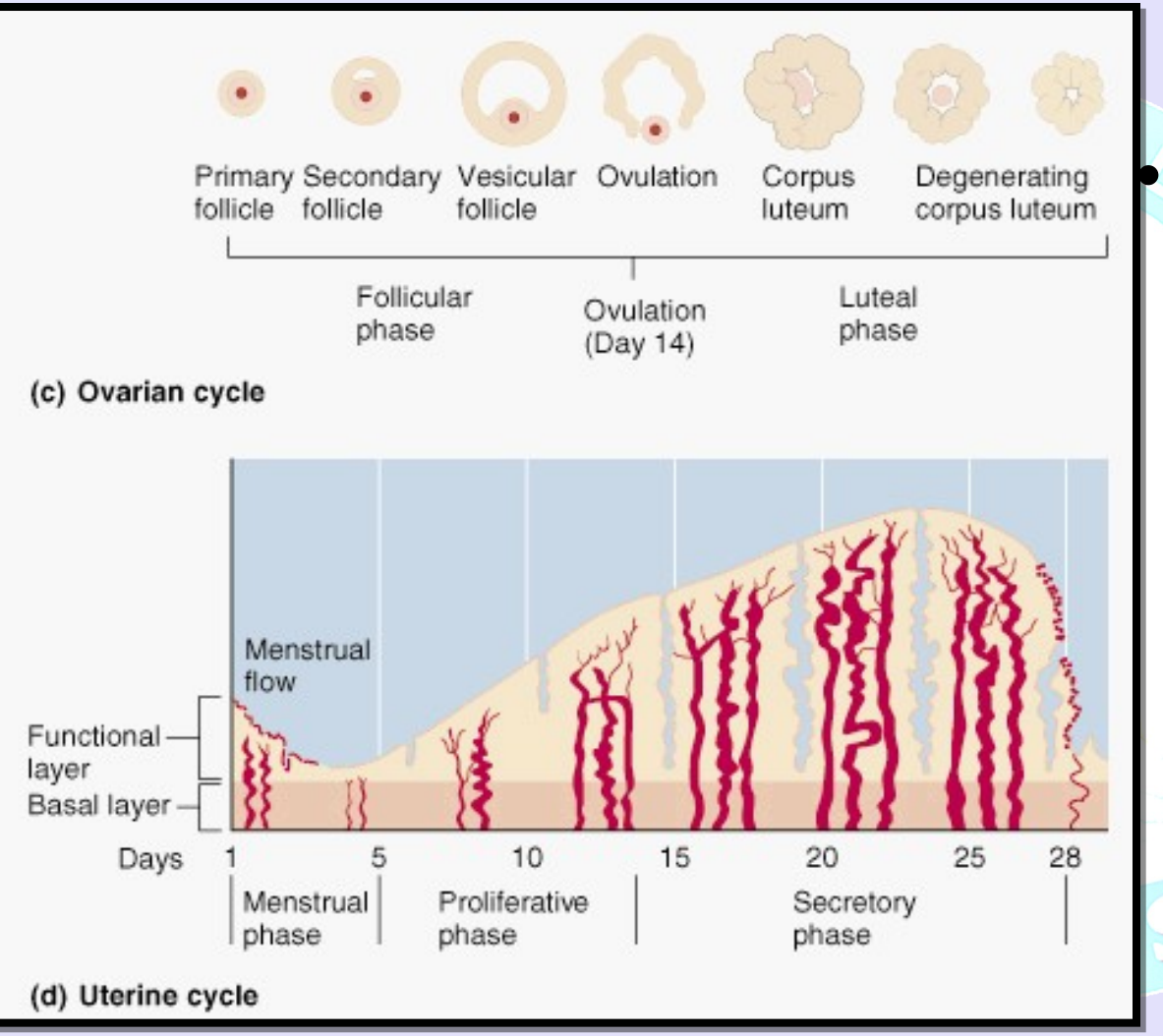
LUTEAL OR SECRETARY PHASE

- Lasts for 13 days
- Under the control of progesterone secreted by corpus luteum
- The glands become wide, tortous & saccular
- They star to secrete glycogen rich material
- The arteries become coiled, the spiral arteries
- Venous network becomes complex & large lacunae developed
- Overall thickness of endometrium increase

INITIATION OF THE MENSTRAUL PHASE

- If fertilization does not occur, then shedding of the endometrium (compact ad spongy) begins, marking the initiation of the menstraul phase .

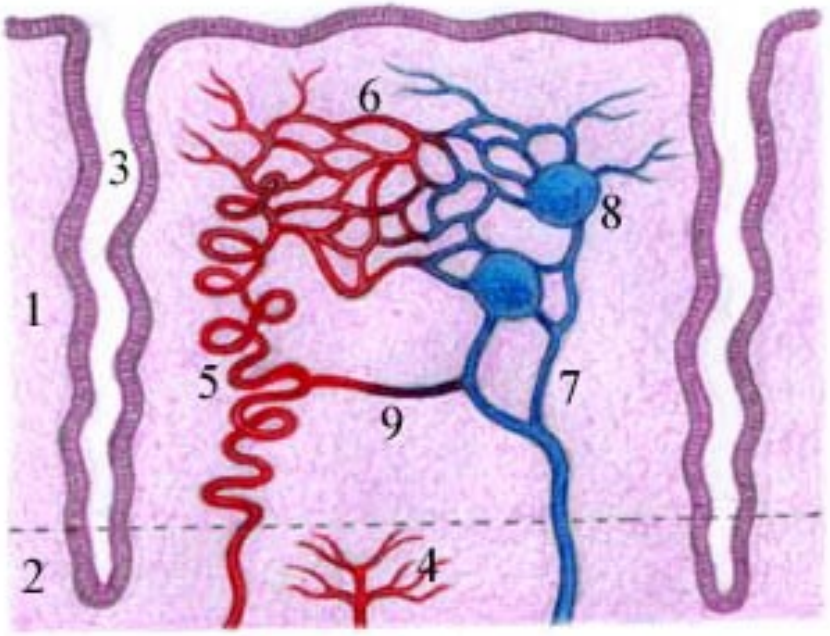




- When the menstrual phase begins , blood escapes from the superficial arteries , and small pieces of stroma and glands break away.

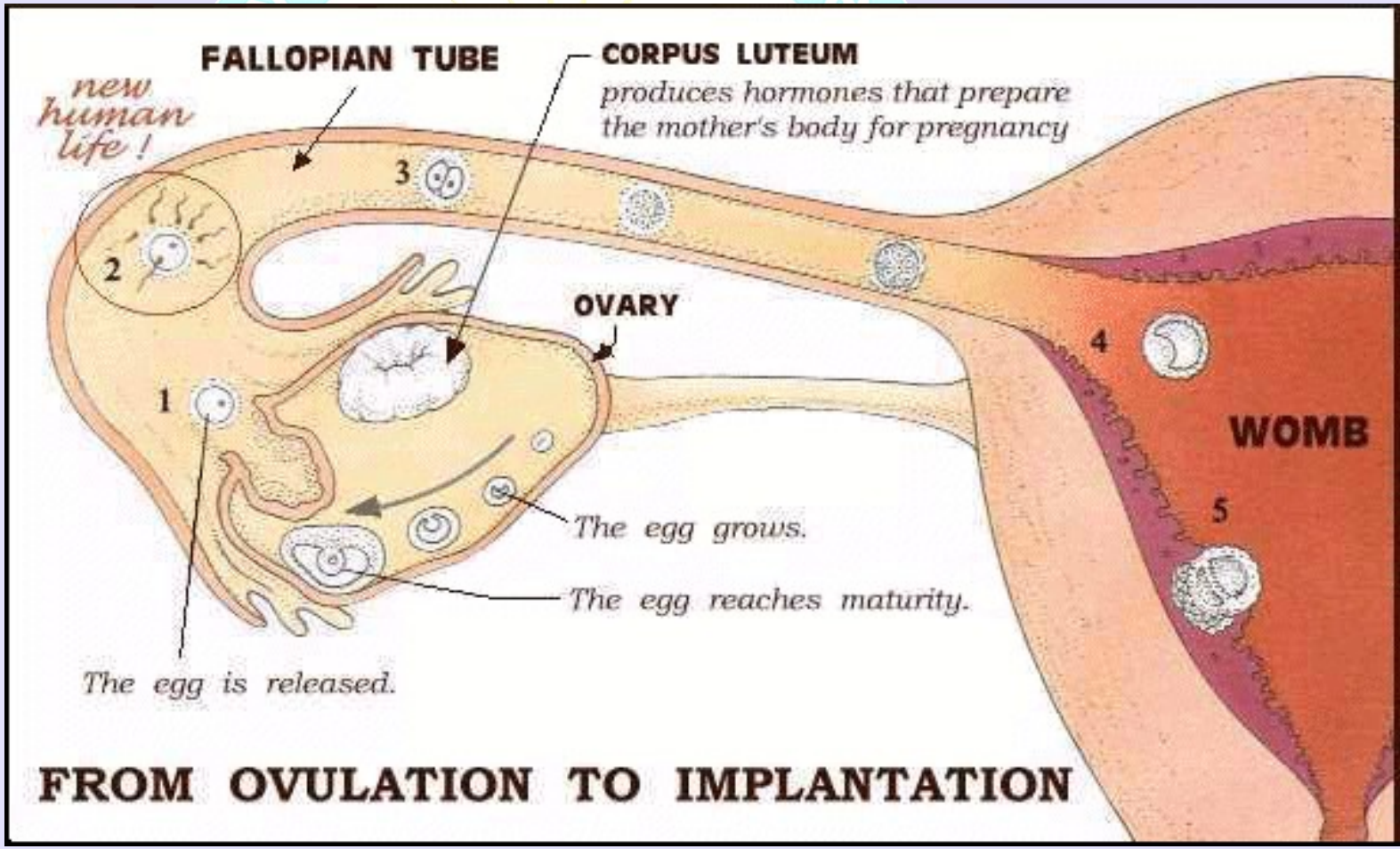
- During the following 3 or 4 days , the compact and spongy layers are expelled from the uterus, and the basal layer is the only part of the endometrium the is retained.

- Basal layer is supplied by its own arteries, the basal arteries , and function as the regenerative layer in the rebuilding of glands and arteries in the proliferative phase.

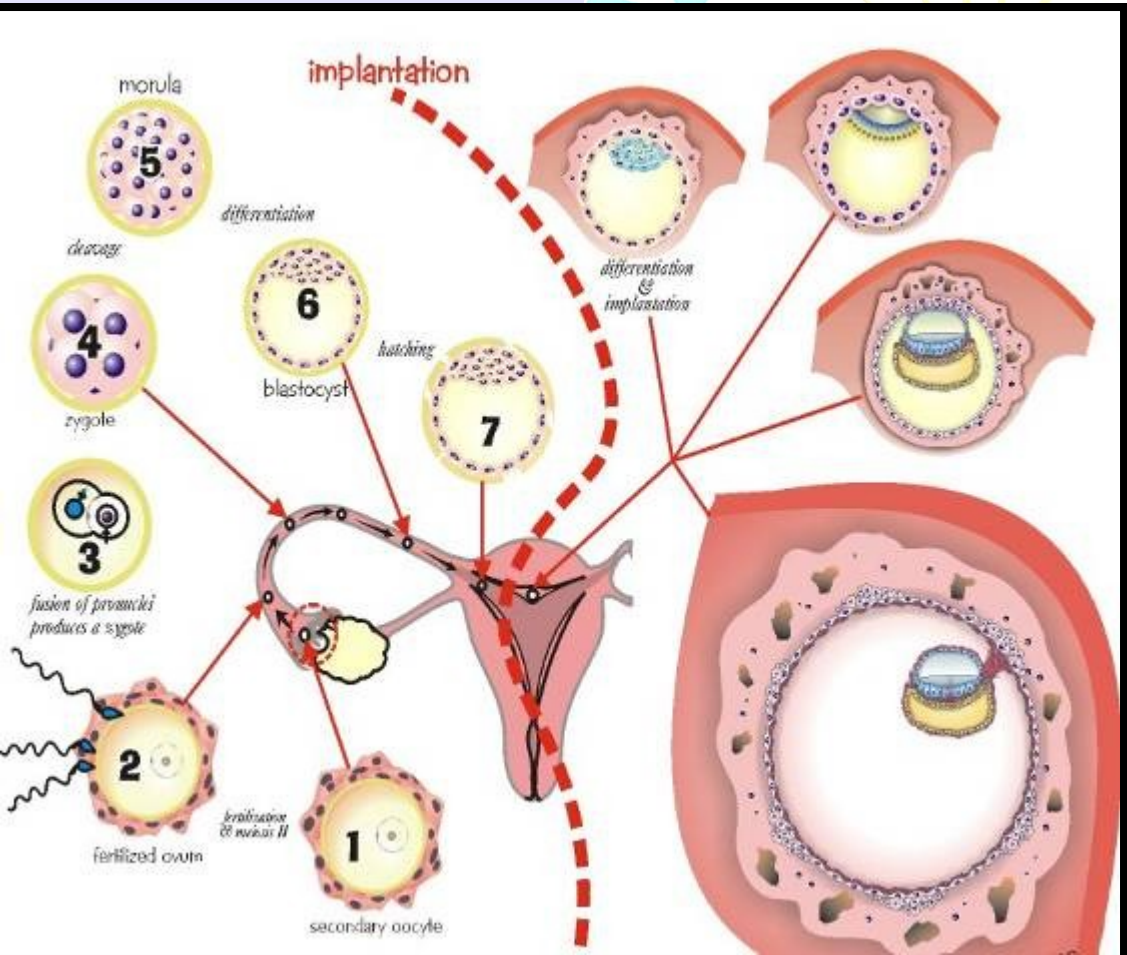


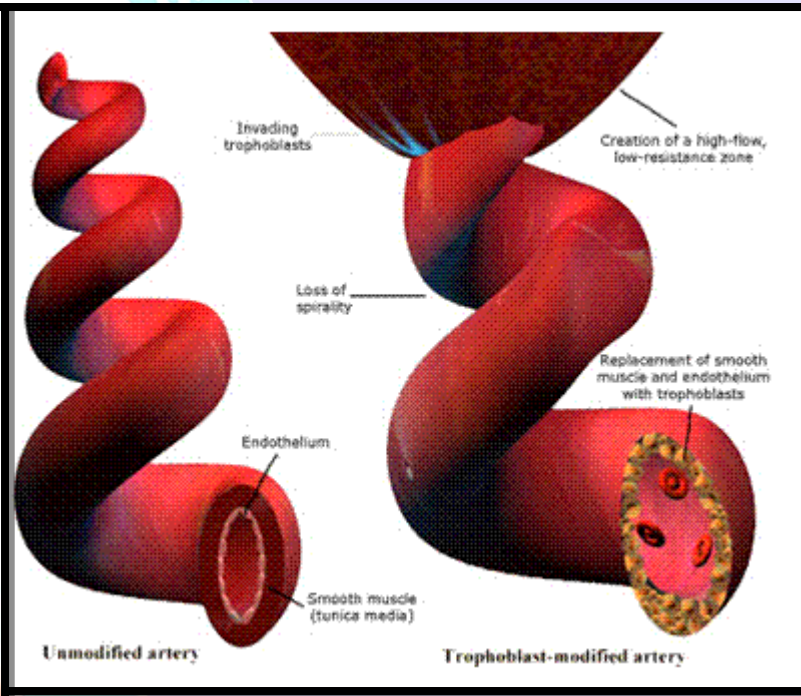
1. Endometrial Functional Layer
2. Endometrial Basal Layer
3. Endometrial Gland
4. Straight Artery
5. Spiral Artery
6. Capillary Plexus
7. Veins
8. Venous Lake
9. Arteriovenous Anastomosis

- If fertilization does occur , then the endometrium assists in implantation and contributes to the formation of the placenta.



- At the time of implantation, the mucosa of the uterus is in the secretory phase, during which time uterine glands and arteries become coiled and the tissue becomes succulent.





spiral arteries
The first half of the cycle

APPLIED ANATOMY

- **Fallopian tube** plays an important role in the mechanical transport and physiological sustenance of the gametes and early conceptus.
- **Compromised tubal damage** can occur after external or internal injury, inhibiting the normal transport of gametes.
- **Pelvic inflammatory disease** is a major clinically unsuspected reason for tubal subfertility.
- **PID** can damage the tube at multiple sites and also predispose to ectopic pregnancy.

References

- **LANGMAN'S - Medical Embryology, Seventh Edition.**


Thank
YOU!